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Introduction of Prelub™ High-Volume Pump

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This service/parts topic introduces a Prelub™ high-volume lubricating oil pump for the QSK45 and QSK60 industrial engines. The high-volume lubricating oil pump reduces the time from approximately 30 to 10 seconds to obtain required oil pressure in the engine. The engine will automatically begin cranking to start after obtaining sufficient oil pressure. This service/parts topic also revises Procedure 007-015, Lubricating Oil Filter Head, and introduces Procedure 007-063, Pre-Lubricating Oil Pump, for the Troubleshooting and Repair Manual, QSK45 and QSK60 Engines, Bulletin 3666261.

The Prelub™ high-volume pump system distributes oil throughout the lubrication system in the engine before starting. The system produces oil pressure in the main lubricating oil rifle and thereby lubricates all friction surfaces in the engine at start up. The risk of starting the engine without sufficient lubricating oil or with dry lubricating oil filters is reduced, and this will lessen engine wear. The automatic system uses a motor-driven pump to supply lubricating oil from the oil pan sump to the main oil rifle prior to starting the engine.

The Prelub™ high-volume pump can be retrofitted to QSK45 and QSK60 industrial engines.

There are three types of Prelub™ systems:

- Integral Prelub™
- Nonintegral Prelub™
- High-volume Prelub™.

The high-volume pump series has increased flow capability and is mandated for engine sizes of 45 liters and larger.

The Prelub™ high-volume pump can be engine- or chassis-mounted on the QSK60 engine with spin-on filters. However, it is **only** available as a remote off-engine installation for the QSK45 and QSK60 engines with Eliminator because of space constraints.

The time taken by the Prelub™ system to generate sufficient pressure is affected by the oil viscosity. The time required to obtain sufficient oil pressure will increase with colder temperatures dependent upon the grade of oil and viscosity. Use arctic weather treatments as advised in the Operation and Maintenance Manual, QSK45 and QSK60 Series Engines, Bulletin 3666410, to maintain acceptable prelubrication times.

Note : Factory installations for the Prelub™ high-volume pump begin with engine serial number first 33149032 for the QSK45 and 33146681 for the QSK60.

Pump and Motor

The chassis-mounted high-volume pump system requires a bracket for mounting the Prelub™ pump. This bracket is included as part of the engine lubrication primer, Part Number 4086437.

On-engine installations require different brackets to mount the pump. Refer to the parts list for a specific application.

The pump mounting location on the chassis **must not** be located higher than 305 mm [12 in] above the engine crankshaft centerline. The pump also **must** be located no farther than 1270 mm [50 in] from the suction port in the oil pan. The pump can be installed in any orientation.

Suction-Side Hose of Prelub™ Plumbing

For optimum performance, the length of the hose **must** be minimized.

The suction-side hose connects the oil pan sump to the Prelub™ pump. For optimum performance, the suction-side hose **must** be minimized. The OEM is responsible for installing the suction hose for the chassis-mounted systems. The Prelub™ high-volume pump system utilizes a number 20 hose with a 1-1/8-inch inside diameter. The OEM cuts the hose to fit the installation. The suction hose can **not** exceed 1270 mm [50 in].

The oil supply port location varies by engine model. Refer to the oil pan option in the sales handbook for the designated Prelub™ port. The oil drain ports **must not** be used.

Clips for the hose **must** be used to avoid rubbing or excessive vibration. The clips are OEM supplied.

Pressure-Side Hose of Prelub™ Plumbing

The pressure-side-hose connects the Prelub™ pump to the engine. The Prelub™ high-volume pump system is supplied with a number 16 hose with a 7/8-inch inside diameter. The pressure-side hose **must** be cut to fit the OEM application. The high-volume pump oil return port **must** have a 20.5-mm [0.81-in] inside diameter (o-ring boss M33).

Installations use **only** high-quality SAE connections and fittings for oil lines. Diameters of the pressure and suction hoses, as well as the length of the suction hose, determine the length of the pressure-side hose.

Maximum hose length of the pressure hose should be no more than four times the length of the suction hose.

The port location for the oil pressure hose will vary by engine model. Acceptable port locations are documented on engine installation diagrams. For every installation the pressure side hose **must** be connected before the engine lubrication filters.

Install the check valve at the location the pressure-side hose is attached to the engine. Oil flows through the valve toward the engine. The valve prevents reverse flow of oil after the engine is started.

⚠ CAUTION ⚠

Make certain the check valve is installed correctly. Incorrect installation can result in Prelub™ motor failure.

Pressure Switch

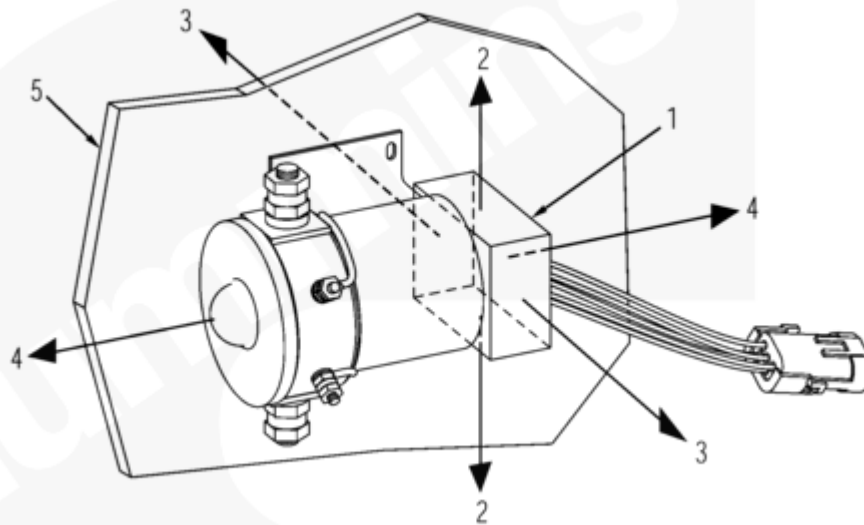
A pressure switch (normally closed) is included. This switch **must** have the oil pressure supply line located to sense oil pressure after it has passed through the full-flow oil filters. The switch has a 2.5 ± 0.5 -psi pressure setting and a two-terminal Packard connection that improves installation of the wiring harness to the switch.

The recommended location is at the main oil rifle at the rear of the block or flywheel housing.

Timer Relay

The timer relay **must** be mounted off the engine and is the responsibility of the OEM. The mounting technique **must** protect the unit from shock, vibration, excessive temperatures, and high-pressure spray.

The relay **must** be mounted to minimize the shock and vibrations along the axis of the relay canister. For most wheeled applications, the timer relay can be mounted with the canister horizontal relay axis at 90 degrees to the crankshaft centerline. Mounting the canister horizontally minimizes the chance of the switch being activated because of vertical shock loading. Mounting the canister at 90 degrees to the direction of travel minimizes the chance of the switch activating because of shock loading from the vehicle.



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Relay Mounting

Table 1, Relay Mounting

Item Number	Description
1	Timer relay
2	Vertical shock load
3	Direction of vehicle travel (end-to-end shock loads)
4	Relay axis
5	Flat mounting surface

The relay **must** also be mounted to avoid heat sources such as mufflers and exhaust manifolds. The relay **must not** be exposed to ambient temperatures over 85°C [185°F]. Verify temperatures are **not** above the listed limit.

⚠ CAUTION ⚠

The relay must be protected from direct contact from high-pressure wash. Moisture can enter the canister and cause damage.

A wire harness for the Prelub™ system is provided. The wire harness will connect the pressure switch and the Prelub™ motor to the timer relay. The OEM is responsible for providing system voltage to the B+ terminal on the engine-mounted Prelub™ motor or the B+ terminal of the chassis-mounted Prelub™ harness. The OEM is also responsible for providing a ground wire from the Prelub™ motor to the batteries. The wiring for the Prelub™ system and cranking motor **must** meet Cummins wiring requirements. Refer to the local authorized distributor for the wiring requirements.

The harness will also include a fuse to protect the Prelub™ circuit. All circuits (except cranking) **must** be protected by a fuse. The normal steady state amperage draw during a Prelub™ cycle is approximately 200 amps, and the maximum amperage draw is approximately 270 amps. Consequently, the Prelub™ harness is sized smaller than the cranking motor harness. However, if the motor fails during the Prelub™ cycle, the motor is capable of drawing a much higher amperage. The smaller Prelub™ harness is more susceptible to damage when compared to the larger cranking harness. The fuse will protect the Prelub™ harness from progressive damage.

It is recommended that the wire harness included with the Prelub™ system be used in the engine installation. However, if the OEM is unable to use the standard harness, the OEM can accept design responsibility and develop a custom harness. If a custom harness is developed, refer to the local authorized distributor for design guidelines.

Additional recommendations unique to the Prelub™ system include:

- Minimum wire size
Prelub™ high-volume pump: Number 0 gauge
- Maximum resistance of main power circuit
High-volume pump system 0.003 Ω
- Wire insulation capability
120°C [248°F]
- Wire insulation material
Thermoset plastic
- Fuse
Prelub™ high-volume pump 400-amp maximum.

The OEM is responsible for wiring from the battery to the B+ terminal on the Prelub™ motor and for the wiring from the ignition switch to the timer logic module or the Prelub™ wiring harness. Refer to the local authorized distributor.

Table 2, Parts Required for QSK60 Engines with Spin-on Filters			
Part Number	Description	Quantity	Comments
4066169	Bracket, pump	1	Required for QSK60 engines with spin-on filters only
4066180	Bracket, pump	2	Required for QSK60 engines with spin-on filters only
3910495	Screw (M12)	2	Required for QSK60 engines with spin-on filters only

3089016	Screw (M10)	2	Required for QSK60 engines with spin-on filters only
3902460	Screw (M10)	4	Required for QSK60 engines with spin-on filters only
4065275	Harness	1	Required for QSK45/60 engines (except KMS and Hitachi customers)
4065465	Switch, pressure	1	Required for all QSK45 and QSK60 engines
4065468	Switch, relay	1	Required for all QSK45 and QSK60 engine (except Hitachi customers)
4065496	Elbow, male union	1	Required for QSK45 and QSK60 engines with spin-on filters and QSK60 engines with new Eliminator, Part Number 4066080
3393258	Elbow, male adapter	1	Required for all QSK45 and QSK60 engines, except: QSK45 KMS, Hitachi customers, WA1200, Komatsu 830E, and 930E haul trucks. (See hoses for these applications at the end of this list.)
4080229	Hose, flexible (supply)	1	Required for QSK60 engines with spin-on filters, except: WA1200, Komatsu 830E, and 930E haul trucks. (See hoses for these applications at the end of this list.)

4080230	Hose, flexible (suction)	1	Required for QSK60 engines with spin-on filters and QSK45 for Liebherr R994 application
4080590	Primer, engine lubricating	1	Required for all QSK60 engines with spin-on filters only
3040815	Plug, threaded	1	Required for all QSK45 and QSK60 engines (required for converting integral pre-lubricating starter motor into a standard starter)
3045782	Plug, threaded	1	Required for all QSK45 and QSK60 engines (required for converting integral prelubricating starter motor into a standard starter)
4007204	Valve, check	1	Required for all QSK45 and QSK60 engines, except the QSK45 for Hitachi customers
4007207	Connector, male	1	Required for all QSK45 and QSK60 engines, except QSK45 for KMS and Hitachi customers
3410392	Elbow, male union	1	Required for all QSK45 and QSK60 engines, except QSK45 for Hitachi customers
4080234	Clip	1	Required for QSK60 engines with spin-on filters only

67950	Clip	1	Required for QSK60 engines with spin-on filters only
3628554	Brace, tube	1	Required for QSK60 engines with spin-on filters only
3093828	Nut	1	Required for QSK60 engines with spin-on filters only
3903990	Screw	1	Required for QSK60 engines with spin-on filters only
3200162	Spacer	1	Required for QSK60 engines with spin-on filters only
3089232	Screw	1	Required for QSK60 engines with spin-on filters only
4080218	Union, male	1	Required for QSK60 engines with spin-on filters only
4085953	Hose, flexible (supply)	1	Required for QSK60 engines in WA1200 application and QSK45 for Liebherr in R994 application
4085954	Hose, flexible (suction)	1	Required for QSK60 engines in WA1200 application only
4080576	Hose, flexible (supply)	1	Required for QSK60 engines in Komatsu 830E and 930E haul trucks only
4080577	Hose, flexible (suction)	1	Required for QSK60 engines in Komatsu 830E and 930E haul trucks only
3021618	Plunger, pressure regulator	1	Required for QSK60 engines with spin-on filter head only
3028552	Body, bypass valve	1	Required for QSK60 engines with spin-on filter head only

3089238	Plug, threaded	1	Required for QSK60 engines with spin-on filter head only
3678611	Plug, threaded	1	Required for QSK60 engines with spin-on filter head only
3678838	Plug, threaded	1	Required for QSK60 engines with spin-on filter head only
3678921	Plug, threaded	2	Required for QSK60 engines with spin-on filter head only
3902460	Screw, hex flange head	4	Required for QSK60 engines with spin-on filter head only
4065919	Head, lubricating oil filter	1	Required for QSK60 engines with spin-on filter head only
4066190	Adapter, filter head	4	Required for QSK60 engines with spin-on filter head only
4066191	Gasket, cover plate	1	Required for QSK60 engines with spin-on filter head only
4066192	Adapter, filter head	4	Required for QSK60 engines with spin-on filter head only
4066194	Plate, cover	1	Required for QSK60 engines with spin-on filter head only
4066195	Spring, compression	1	Required for QSK60 engines with spin-on filter head only

**Table 3, Parts
Required on QSK60
Engines with
Eliminator**

Part Number	Description	Quantity	Comments
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Table 3, Parts Required on QSK60 Engines with Eliminator			
4065275	Harness	1	Required for all QSK45 and QSK60 engines (except KMS and Hitachi customers)
4065465	Switch, pressure	1	Required for all QSK45 and QSK60 engines
4065468	Switch, relay	1	Required for all QSK45 and QSK60 engines (except Hitachi customers)
4065496	Elbow, male union	1	Required for all QSK45 and QSK60 engines with spin-on filters and QSK60 engines with new Eliminator, Part Number 4066080
4065549	Elbow, male union	1	Required for QSK60 engines with Eliminator Part Number 3410860
4065547	Elbow, male union	1	Required for QSK60 engines with Eliminator, Part Number 3410860
4085936	Hose, flexible (supply)	1	Required for QSK60 engines with Eliminator (except KMS customers) and all QSK45 engines (except KMS, Liebherr, and Hitachi customers)

Table 3, Parts Required on QSK60 Engines with Eliminator			
4085935	Hose, flexible (suction)	1	Required for QSK60 engines with Eliminator (except KMS customers) and all QSK45 engines (except KMS, Liebherr, and Hitachi customers)
4065714	Elbow, male union	1	Required for QSK60 engines with Eliminator and all QSK45 engines except KMS and Hitachi customers
3393258	Elbow, male adapter	1	Required for all QSK45 and QSK60 engines, except QSK45 KMS and Hitachi customers
4086437	Primer, engine lubrication	1	Required for QSK45 and QSK60 engines with off-engine high-volume pump (HVP) installations (mounting bracket included)
3040815	Plug, threaded	1	Required for all QSK45 and QSK60 engines (required for converting integral prelubricating starter motor into a standard starter)

Table 3, Parts Required on QSK60 Engines with Eliminator			
3045782	Plug, threaded	1	Required for all QSK45 and QSK60 engines (required for converting integral prelubricating starter motor into a standard starter)
4007204	Valve, check	1	Required for all QSK45 and QSK60 engines, except QSK45 for Hitachi customers
4007207	Connector, male	1	Required for all QSK45 and QSK60 engines, except QSK45 for KMS and Hitachi customers
3410392	Elbow, male union	1	Required for QSK45 and QSK60 engines, except QSK45 for Hitachi customers
4085953	Hose, flexible (supply)	1	Required for QSK60 engines in WA1200 application and QSK45 for Liebherr in R994 application
4085954	Hose, flexible (suction)	1	Required for QSK60 engines in WA1200 application only
4080576	Hose, flexible (supply)	1	Required for QSK60 engines in Komatsu 830E and 930E haul trucks only
4080577	Hose, flexible (suction)	1	Required for QSK60 engines in Komatsu 830E and 930E haul trucks only

Table 4, Parts Required on QSK45 Engines with Spin-On Filter Heads

Part Number	Description	Quantity	Comments
4065275	Harness	1	Required for all QSK45 and QSK60 engines (except KMS and Hitachi customers)
4065465	Switch, pressure	1	Required for all QSK45 and QSK60 engines
4065468	Switch, relay	1	Required for all QSK45 and QSK60 engines (except Hitachi customers)
4065496	Elbow, male union	1	Required for QSK45 and QSK60 engines with spin-on filters and QSK60 engines with a new Eliminator, Part Number 4066080
4085936	Hose, flexible (supply)	1	Required for QSK60 engines with Eliminator (except KMS customers) and all QSK45 engines (except KMS, Liebherr, and Hitachi customers)
4085935	Hose, flexible (suction)	1	Required for QSK60 engines with Eliminator (except KMS customers) and all QSK45 engines (except KMS, Liebherr, and Hitachi customers)

Table 4, Parts Required on QSK45 Engines with Spin-On Filter Heads			
Part Number	Description	Quantity	Comments
4065714	Elbow, male union	1	Required for QSK60 engines with Eliminator and all QSK45 engines except KMS and Hitachi customers
3393258	Elbow, male adapter	1	Required for all QSK45 and QSK60 engines, except QSK45 KMS and Hitachi customers
4080230	Hose, flexible (suction)	1	Required for QSK60 engines with spin-on filters and QSK45 for Liebherr R994 application
4086437	Primer, engine lubricating	1	Required for QSK45 and QSK60 engines with off-engine high-volume pump installations (mounting bracket included)
3040815	Plug, threaded	1	Required for all QSK45 and QSK60 engines (required for converting integral pre-lubricating starter motor into a standard starter)
3045782	Plug, threaded	1	Required for all QSK45 and QSK60 engines (required for converting integral pre-lubricating starter motor into a standard starter)

Table 4, Parts Required on QSK45 Engines with Spin-On Filter Heads			
Part Number	Description	Quantity	Comments
4007204	Valve, check	1	Required for all QSK45 and QSK60 engines, except QSK45 for Hitachi customers
4007207	Connector, male	1	Required for all QSK45 and QSK60 engines, except QSK45 for KMS and Hitachi customers
3410392	Elbow, male union	1	Required for all QSK45 and QSK60 engines, except QSK45 for Hitachi customers
4080216	Head, lubricating oil filter	1	Required for QSK45 engines with spin-on filters
4085953	Hose, flexible (supply)	1	Required for QSK60 engines in WA1200 application and QSK45 for Liebherr in R994 application

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